NETL's Carbon Capture Retrofit Databases

NATIONAL ENERGY TECHNOLOGY LABORATORY

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Agenda

- Overview
- Design Assumptions
- Financial Methodology
- Default Cost of Capture Results
- Industrial Carbon Capture Retrofit Database
- Discussion

Overview

- NETL published three carbon capture studies in 2022–2023.
- *Cost of Capturing CO_2 from Industrial Sources** (Industrial Report).
  - Quantifies the performance of CO_2 compression and solvent-based capture systems and their greenfield and retrofit costs of CO_2 capture when applied at nine representative industrial sources.
  - Representative plant sizes and CO_2 available for capture from each industrial plant were chosen based on literature review, vendor input, and other NETL studies, where appropriate.
  - This study does not characterize reference industrial plants beyond the flowrate and stream characteristics of the available CO_2 stream. The production costs of each plant's product before and after retrofit are not considered.
- *Cost and Performance of Retrofitting NGCC Units for Carbon Capture.***
  - Quantifies the cost and performance of retrofitting natural gas combined cycle (NGCC) power plants with solvent-based post-combustion carbon capture.

**Cost of Capturing CO_2 from Industrial Sources* is publicly accessible via https://www.osti.gov/biblio/1887586*
***Cost and Performance Baseline of Retrofitting NGCC Units for Carbon Capture* is publicly accessible via https://www.osti.gov/biblio/1961845*

Overview (continued)

- *Eliminating the Derate of Carbon Capture Retrofits.**
  - Quantifies options for existing pulverized coal (PC) boilers to reduce the effects caused by steam and power requirements of retrofit solvent-based capture systems and the resulting derated output of the existing plant.
  - Allows for addition of supplemental heat and power provisions in support of CO_2 capture (e.g., natural gas (NG)-fired boiler for solvent-regeneration heating needs and NG simple-cycle for combined heat and power supply).
- To accompany those studies, Microsoft Excel-based tools were developed to provide public users with the ability to perform high-level analysis on the incremental costs for retrofitting industrial plants and electricity generation units with CO_2 capture and/or compression systems.

**Eliminating the Derate of Carbon Capture Retrofits* is publicly accessible via https://www.osti.gov/biblio/1510790

Overview (continued)

- Carbon Capture Retrofit Databases (CCRDs).*
  - Publicly available spreadsheet tools to calculate high-level estimates of the cost of CO_2 capture (COC) for a set of facilities or for entire sectors.
  - Enables scenario analysis based on assumptions from respective NETL analyses alongside manipulatable user inputs.
  - Quickly determines the cost impacts of changing inputs such as CO_2 capture rate, financing assumptions, utility pricing (e.g., natural gas, purchased power), and operating conditions (e.g., cooling type, heat supply, and capacity factor).

A	B	C	D	E	F	G	
User Inputs and Constant							
Input Desired Scenarios t							
Parameter	**Units**	**Value**					
		Ammonia	**Cement**	**Ethanol**	**Hydrogen**	**Natural Gas**	
CO_2 Capture Rate (Source 1)	Choose option	Default	Default	Default	Default	Default	Ca
	%	100%	90%	100%	99%	100%	Th
CO_2 Capture Rate (Source 2)	Choose option	Default	N/A	N/A	N/A	N/A	Ca
	%	90%	N/A	N/A	N/A	N/A	Th
Capacity Factor	Choose Option	Default	Default	Default	Default	Default	De
	%	84%	79%	84%	87%	61%	Ra
Retrofit Unit Capacity Applicability Limit	Choose Option	Default	Default	Default	Default	Default	Al
	TPD-CO_2	0	0	0	0	0	ra
Retrofit Cost Factor	Choose Option	Default	Default	Default	Default	Default	M
		1.05	1.05	1.05	1.05	1.05	
Capital Charge Factor	Choose Option	Default	Default	Default	Default	Default	An
	Choose Option	30-year	30-year	30-year	30-year	30-year	co
		0.055	0.054	0.070	0.046	0.060	va
Advanced Options							
Maximum CO_2 Capture Rate Per Train	Choose option	Default	Default	Default	Default	Default	If
	TPD	21,867	27,430	122,453	54,516	121,229	
Cost Year Basis	Choose Option	Default	Default	Default	Default	Default	Sc
	Year	2018	2018	2018	2018	2018	av
Cooling preference	Choose Option	State	State	State	State	State	St
Purchased Electricity Price	Choose Option	State	State	State	State	State	Pri
	$/MWh	Varies by State	Varies by State	Varies by State	Varies by State	Varies by State	th
Purchased Natural Gas Price	Choose Option	State	State	N/A	State	N/A	Pri
	$/MMBtu	Varies by State	Varies by State	N/A	Varies by State	N/A	ea
Additional Auxiliary Load Penalty	Choose Option	None	None	None	None	None	Th
	kW/TPD-CO_2	N/A	N/A	N/A	N/A	N/A	
CO_2 Transport and Storage Costs	Choose Option	Default	Default	Default	Default	Default	Th
	$/tonne captured	10	10	10	10	10	Th
Include Natural Gas Boiler Emissions in Capture Feed	Choose Option	Yes	Yes	N/A	Yes	N/A	If

Industrial CCRD is publicly accessible via https://www.osti.gov/biblio/1887588
PC CCRD is publicly accessible via https://www.netl.doe.gov/energy-analysis/details?id=e7e822ff-18ac-4bc6-a052-0be3521b8789
NGCC CCRD is publicly accessible via https://www.netl.doe.gov/energy-analysis/details?id=428b6b31-5a0b-4581-a7b0-12da7a301878

Overview

Limitations

- The cost and performance results presented in the CCRDs are based on assumptions necessitated by limitations of available data. These uncertainties compound those of the reference cost and performance data used as the basis for developing the reported results.

- The CCRDs are not meant to provide actual CO_2 capture costs for any plant or sector—too many site-specific variables make such accuracy infeasible—but serve as a valuable tool for comparative analysis between plants within an industry or for comparing one industry to another (e.g., comparing capture costs in the ammonia sector versus the cement sector).

Design Assumptions

- In 2021, Shell provided quotes for two amine-based capture systems.
  - CANSOLV® post-combustion capture: quote for Cement and Iron/Steel (industrial) applications, quote for PC applications, and quote for NGCC applications.
  - ADIP-Ultra® pre-combustion capture: quote for Refinery Hydrogen (industrial) application.

  Each system was quoted at 90, 95, and 99% capture rates, except for NGCC, which was quoted at 90, 95, and 97% capture.
- Other vendor quotes for an NG boiler, NG simple-cycle combustion turbine, and reciprocating CO_2 compressor were obtained from various original equipment manufacturers in support of recent NETL studies, including those used to develop the CCRDs.
- Retrofit costs were estimated for all cases by applying a factor to the total plant cost (TPC) of incremental capital equipment required to support CO_2 capture.
  - This includes, but is not limited to, items such as the solvent-based capture system; other gas clean-up unit operations; CO_2 compression, cooling, and drying equipment; and ancillary equipment and facilities.

Design Assumptions (continued)

- Aspen Plus® simulations of each representative capture system were created to provide process parameters used to scale capital costs and estimate annual operating and maintenance (O&M) costs for consumables, waste disposal, fuel, power, and co-product costs, as applicable.
  - For industrial cases, only the CO_2 capture systems—and the NG boiler used to raise steam for solvent regeneration, where necessary—were modeled based on assumptions made to characterize the emissions streams leaving each representative industrial plant.
    - Industrial capture system auxiliary load demands are met with power purchased from the grid.
  - For NGCC and PC cases, the entire power generation plant is modeled to account for derate due to the additional auxiliary load demands of the capture system and steam extraction required for solvent regeneration heating needs.
- Capital costs were scaled per guidance in the latest revision of NETL's *QGESS Capital Cost Scaling Methodology: Revision 4 Report*.*
- Per NETL's *QGESS Cost Estimation Methodology for NETL Assessments of Power Plant Performance,** some O&M costs are calculated based on TPC (e.g., maintenance labor, property taxes and insurance, and maintenance materials).

**NETL maintains a library of Quality Guidelines for Energy System Studies (QGESS) that is used to guide technoeconomic analyses. The QGESS documents are referenced within the report; the CCRD tool and are publicly accessible via https://netl.doe.gov/node/7513*

Financial Methodology

Case-Specific Financial Parameters

- NETL developed industry-specific financial assumptions for each study.
- The resulting capital charge factors (CCFs) are manipulatable user inputs to demonstrate the effects of market volatility on the estimated COCs.

Default Financial Assumptions for Carbon Capture Cases										
Applicable Case(s)	**Ammonia**	**Ethylene Oxide**	**Ethanol**	**NGP**	**CTL/GTL**	**Refinery Hydrogen**	**Cement**	**Iron & Steel**	**PC Retrofit**	**NGCC Retrofit**
CCF = TASC/TOC * FCR	**5.51%**	**4.74%**	**6.96%**	**6.05%**	**7.71%**	**4.55%**	**5.35%**	**7.53%**	**7.73%**	
FCR	5.33%	4.63%	6.64%	5.82%	7.32%	4.39%	5.08%	6.90%	7.07%	
TASC/TOC	1.035	1.025	1.047	1.039	1.054	1.036	1.054	1.091	1.093	
Debt/Equity	54/46	48/52	36/64	43/57	32/68	33/67	42/58	39/61	55/45	
Return on Equity	1.50%	0.04%	4.51%	2.96%	5.54%	0.41%	1.42%	5.02%	7.84%	
Pre-tax WACC	3.46%	2.49%	4.74%	3.90%	5.41%	0.87%	2.99%	5.07%	5.15%	
Payback Period	30-year operational period									
Interest on Debt	5.15%								2.94%	
Capital Expenditure Period	1 year					3 years				
Capital Distribution	1st year – 100%					1st year – 10%; 2nd year – 60%; 3rd year – 30 %				

NGP = NG processing; CTL/GTL = coal to liquids/gas to liquids; TASC = total as-spent cost; TOC = total overnight cost; FCR = fixed charge rate; WACC = weighted average cost of capital; Note: All values represent real dollars

Financial Methodology

Calculating the COC for Industrial Cases

- The COC for industrial cases, excluding transport and storage (T&S), is calculated using the equation below, where T&S costs would be an additive cost if included.

$$\left(\frac{\$}{tonne\ CO_2}\right) = \frac{TOC * CCF + FOM + VOM + PF + PP}{tonnes\ CO_2\ captured\ per\ year}$$

- Where:
  - TOC – Total overnight costs of all equipment added to support capture application.
  - CCF – Capital charge factor based on industry-specific financial assumptions.
  - FOM – Annual fixed O&M costs.
  - VOM – Annual variable O&M costs.
  - PF – Purchased NG fuel.
  - PP – Purchased power.

Financial Methodology

Calculating the COC for Power Cases

- The COC for power cases, excluding T&S, is calculated using the equation below, where T&S costs would be an additive cost if included.

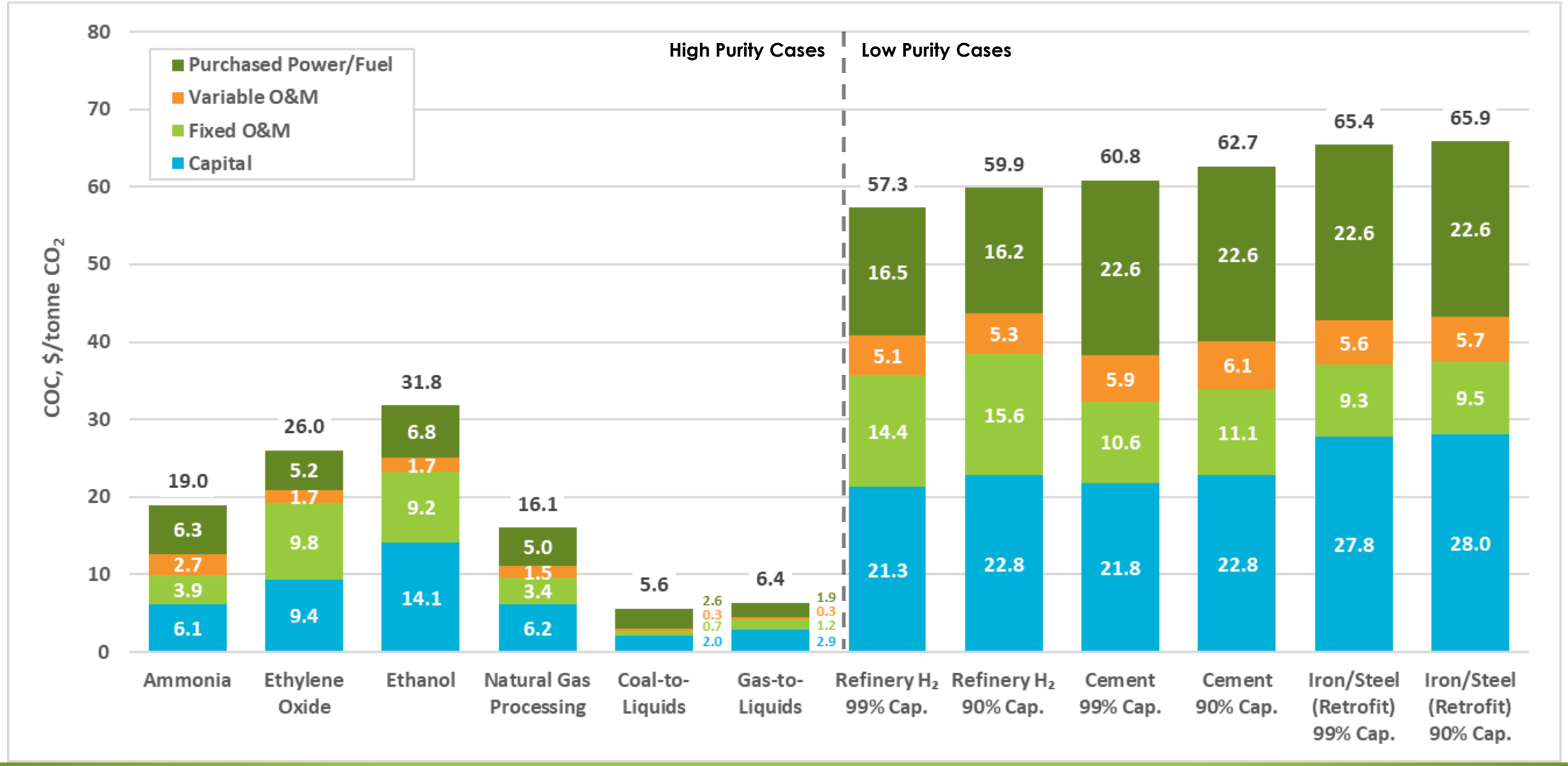
$$\left(\frac{\$}{tonne\ CO_2}\right) = \frac{(COE + MPC - TS - IFOM) * (PRC - TPL)}{tonnes\ CO_2\ captured\ per\ hour}$$

- Where:
  - COE – Cost of electricity including T&S, $/megawatt-hour (MWh).
  - MPC – Makeup/excess power cost, $/MWh.
    - Accounts for the cost of derate to the existing plant gross output, or in cases where NG simple cycle power and heat generation are added, a credit when power is produced over the existing plant gross output less total auxiliary load demand.
  - TS – CO_2 T&S costs, $/MWh.
    - T&S costs are estimated based on guidance in NETL's *QGESS Carbon Dioxide Transport and Storage Costs in NETL Studies** for default cases but is a manipulatable user input in the CCRDs.
  - IFOM – Incremental change in existing plant fixed O&M, $/MWh.
  - PRC – Pre-retrofit capacity, megawatt (MW).
  - TPL – Parasitic retrofit load, MW.

**NETL maintains a library of QGESS that is used to guide technoeconomic analyses. The QGESS documents are referenced within the report; the CCRD tool and are publicly accessible via https://netl.doe.gov/node/7513*

Default Cost of Capture Results

COC for Industrial Cases (Dec. 2018 USD)

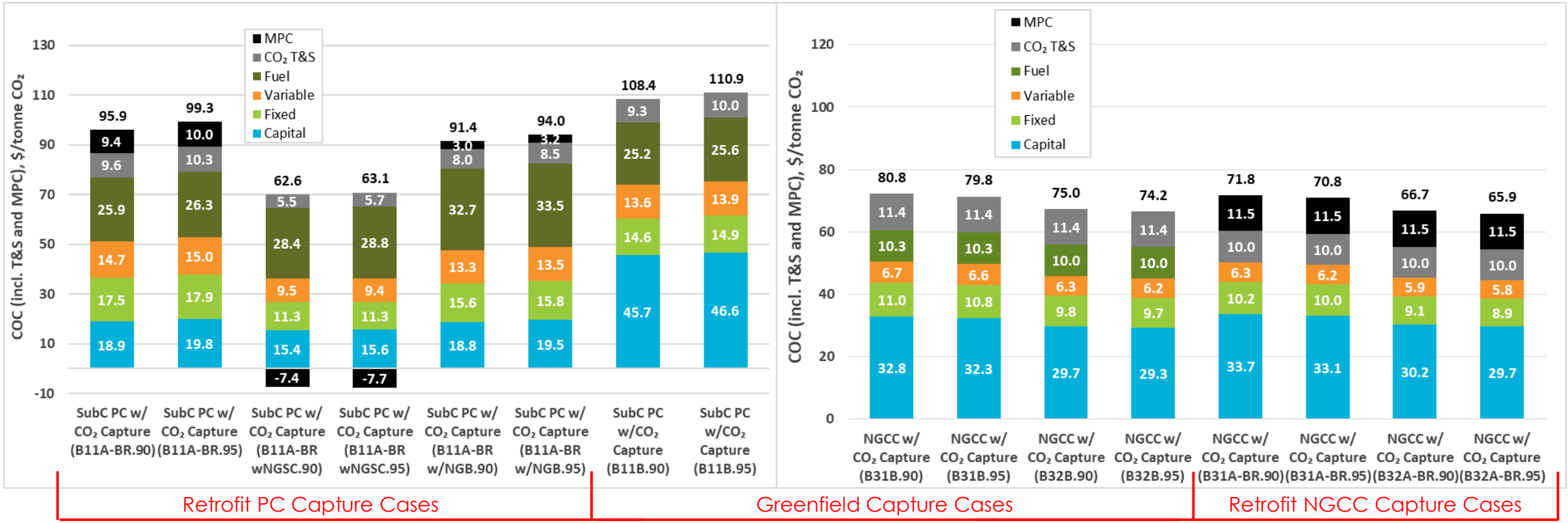


Default Cost of Capture Results

COC (includes T&S and MPC) for Power Cases (Dec. 2018 USD)

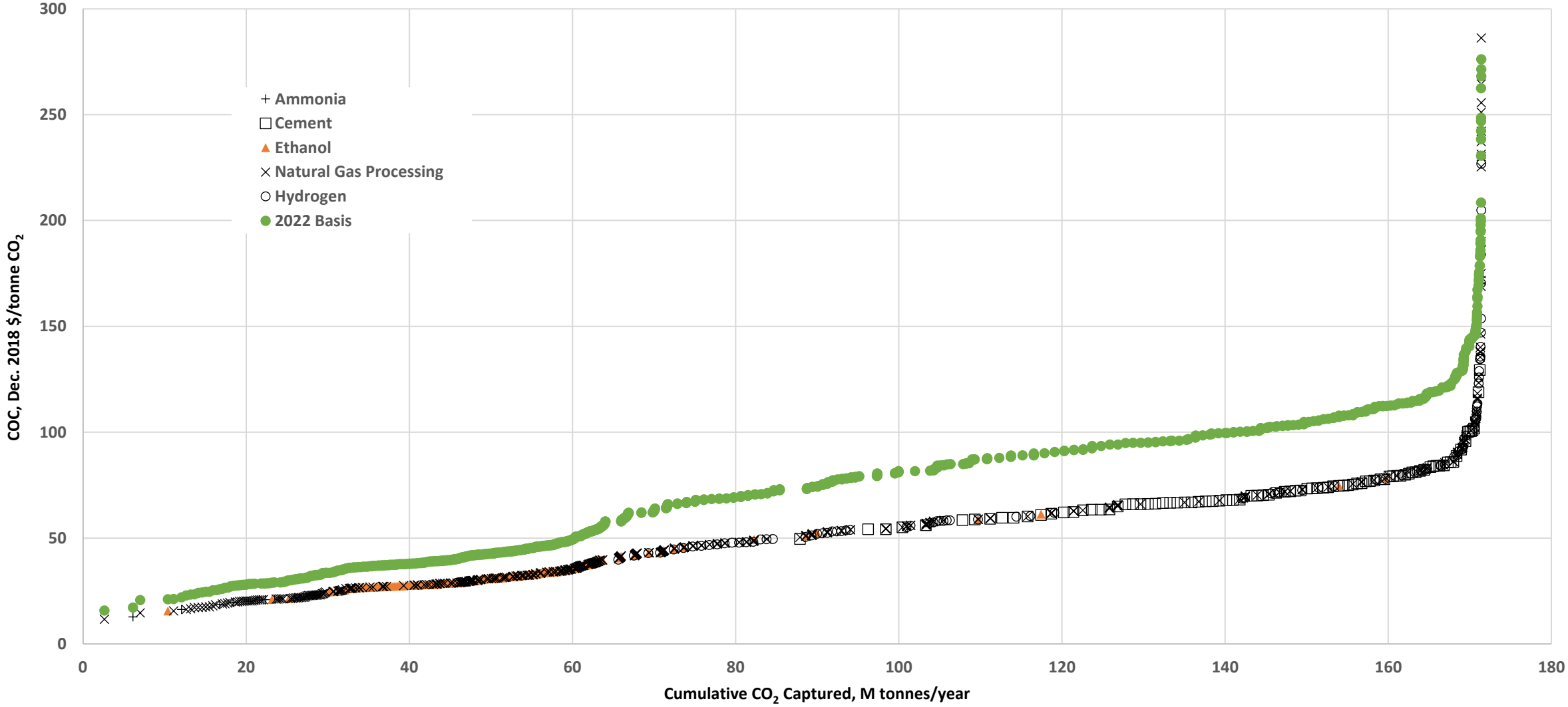
PC Cases

NGCC Cases



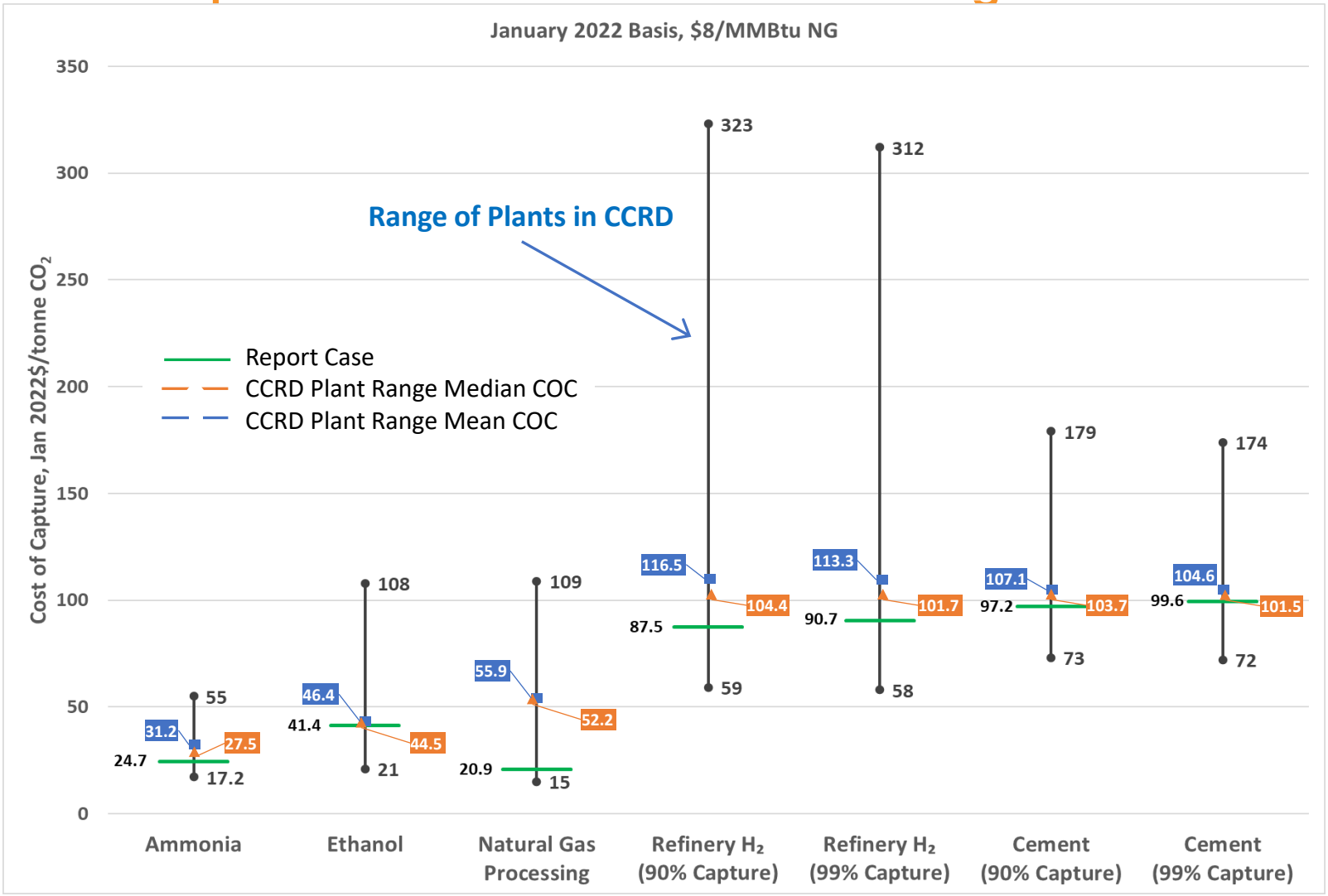
Industrial Carbon Capture Retrofit Database

CCRD CO_2 Supply Curve Using Industrial Report Parameters



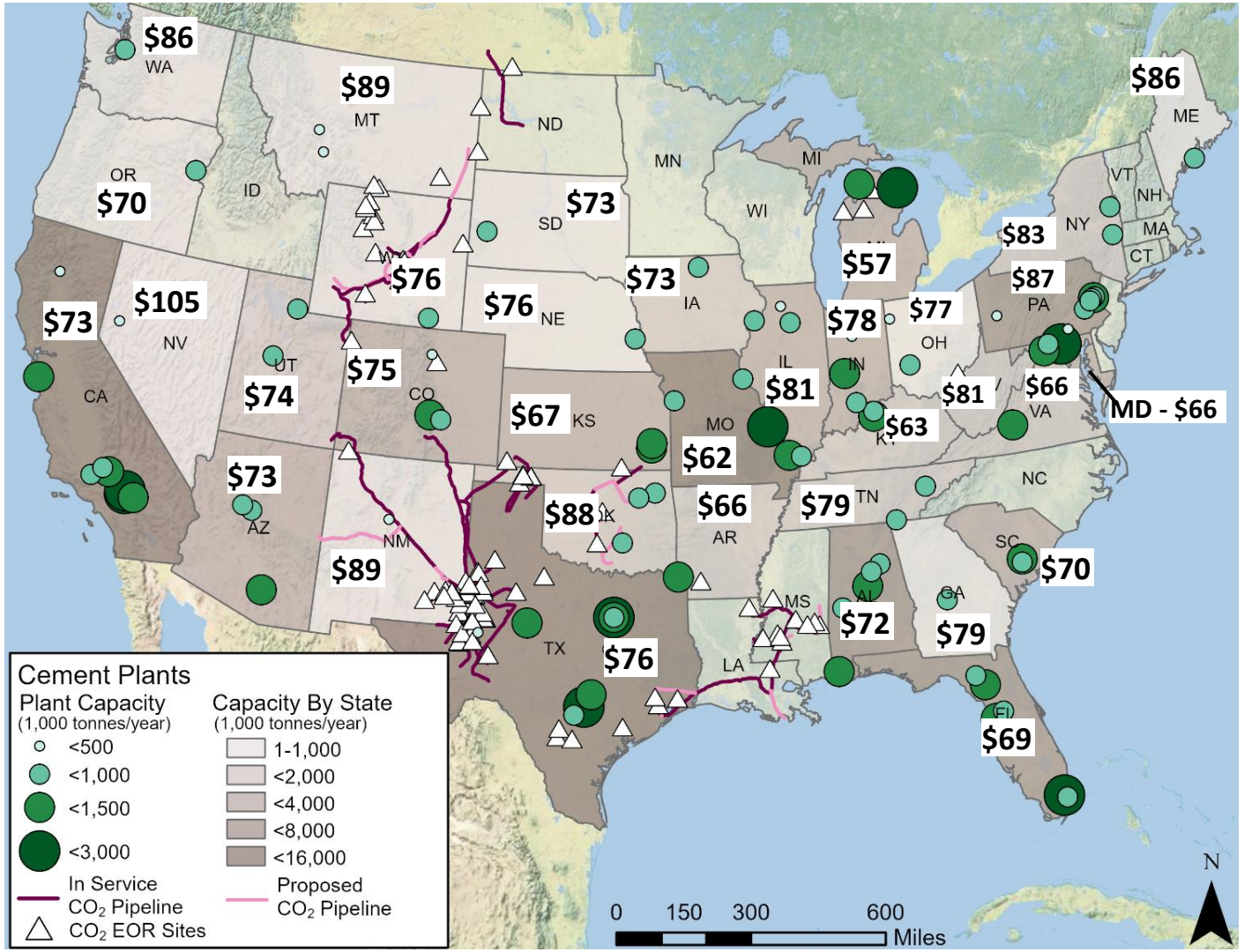
Industrial Carbon Capture Retrofit Database

Comparison of Industrial Report Cases With Plant Database Range From CCRD



Industrial Carbon Capture Retrofit Database

Cement Plants by State – Industrial Report Parameters



Source: NETL

Questions/ Comments

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Content Links

Report: Cost of Capturing CO_2 from Industrial Sources, 2022
https://www.osti.gov/biblio/1887586

Tool: Industrial CCRD, 2022
https://www.osti.gov/biblio/1887588

Report: Eliminating the Derate of Carbon Capture Retrofits, 2023
https://www.osti.gov/biblio/1968037

Tool: PC CCRD, 2023
https://www.netl.doe.gov/energy-analysis/details?id=e7e822ff-18ac-4bc6-a052-0be3521b8789

Report: Cost and Performance of Retrofitting NGCC Units for Carbon Capture, 2023
https://www.osti.gov/biblio/1961845

Tool: NGCC CCRD, 2023
https://www.netl.doe.gov/energy-analysis/details?id=428b6b31-5a0b-4581-a7b0-12da7a301878

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